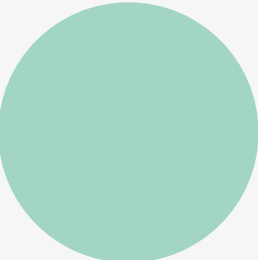


Elephant Herding Optimization



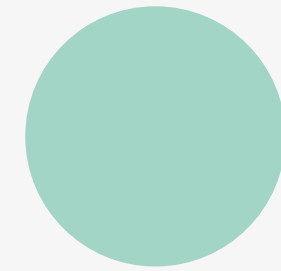
Planning &
Optimization
Techniques

Team Members



- Mohammed Emad Mohammed Elsafty
- Mohamed Magdy Abdel-Qader Mabrouk
- Giovanni Aaron Rawi Shafik
- Nesma Hesham Fathy
- Malak Ahmed Samih
- Menna Haitham Salah

Introduction



Optimization problems appear in many real-world applications such as engineering design, machine learning, scheduling, routing, and resource allocation.

The Elephant Herding Optimization (EHO) is a swarm intelligence-based metaheuristic optimization technique inspired by the social behaviour of elephant herds in nature.

In the wild, elephants live in groups called clans, led by a matriarch. The algorithm balances exploration and exploitation to solve complex, high-dimensional optimization problems.



EHO

Definition



Elephant Herding Optimization (EHO) is a population-based metaheuristic optimization algorithm originally proposed by Wang, Deb, and Coelho in 2015.

It simulates the complex and highly organized social structure of elephant herds to find optimal or near-optimal solutions for challenging real-world problems.

EHO is particularly effective for high-dimensional, nonlinear, multimodal, and derivative-free optimization tasks where traditional methods struggle.

Natural Behavior



Elephants live in a highly organized matriarchal society:

- Elephants form small family groups called clans.
- Each clan is led by the oldest and most experienced female → the matriarch.
- Clan members follow the matriarch closely to ensure group protection and efficient foraging (exploitation).
- The matriarch moves toward the clan center to maintain unity and balance.

Separating behavior:

When young males reach maturity, they leave the clan .. this prevents inbreeding and introduces diversity (exploration).

Key Parameters



Number of clans

Population size (N)

Search space bounds:
[El_min, El_max]

α (alpha $\in [0,1)$)

β (beta $\in [0,1)$)

Random numbers: R
 $\in [0,1]$ and rand(0,1)
for diversity

Algorithm Steps (1/3)



1. Solution Representation

Each elephant is a candidate solution in D-dimensional space:

$$El_i = (x_{i1}, x_{i2}, \dots, x_{iD})$$

2. Initialization

$$El_{i,j} = El_{min,j} + rand(0, 1) \times (El_{max,j} - El_{min,j})$$

3. Fitness Evaluation

Compute $f(El_i)$ for each elephant (minimize or maximize depending on the problem).

Algorithm Steps (2/3)



Clan Updating Operator (Exploitation)

Updating normal elephants:

$$El_{new} = El_{old} + \alpha \times (El_{best} - El_{old}) \times R$$

Where:

- (El_{old}) : current position
- (El_{new}) : updated position
- (El_{best}) : matriarch (best elephant in the clan)
- (α in $[0,1]$) : scaling factor controlling influence
- (R in $[0,1]$) : random number to increase diversity

Matriarch Updating Operator

$$El_{best}^{new} = \beta \times El_{center}$$

Algorithm Steps (3/3)



Clan Center Calculation

$$El_{center} = \frac{1}{n_{clan}} \sum_{j=1}^{n_{clan}} El_j$$

Separating Operator (Exploration)

$$El_{worst} = El_{min} + rand(0, 1) \times (El_{max} - El_{min})$$

Pseudocode



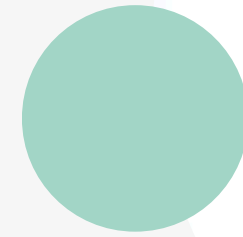
Algorithm: Elephant Herding Optimization (EHO)



1. Begin
2. Initialize population P randomly within $[x_{\min}, x_{\max}]$
3. Set MaxGen, α , β , number of clans
4. $T = 1$
5. While $T \leq \text{MaxGen}$ do
6. Evaluate fitness
7. Sort population
8. Divide into clans
9. For each clan:
10. Update non-matriarch elephants using Eq. (1)
11. Update matriarch using Eq. (2) via Eq. (3)
12. Replace worst elephant in each clan using Eq. (4)
13. $T = T + 1$
14. End while
15. Return best solution End

How It Works

Step-by-Step Implementation



- Initialization (parameters + random population)
- Fitness evaluation & sorting
- Divide into clans + identify matriarchs
- Clan Updating
- Separating Operator
- Update population & recalculate fitness
- Repeat until MaxGen or convergence
- Return the global best solution

Conclusion



Elephant Herding Optimization (EHO) is a powerful, simple, and effective metaheuristic algorithm.

It successfully balances exploration (separating operator) and exploitation (clan updating operator).

Advantages:

- Few control parameters
- Strong global search capability
- Good convergence speed
- Competitive with GA, PSO, and ACO

EHO is suitable for engineering, machine learning, scheduling, routing, and many other complex optimization problems.





Thank you!

